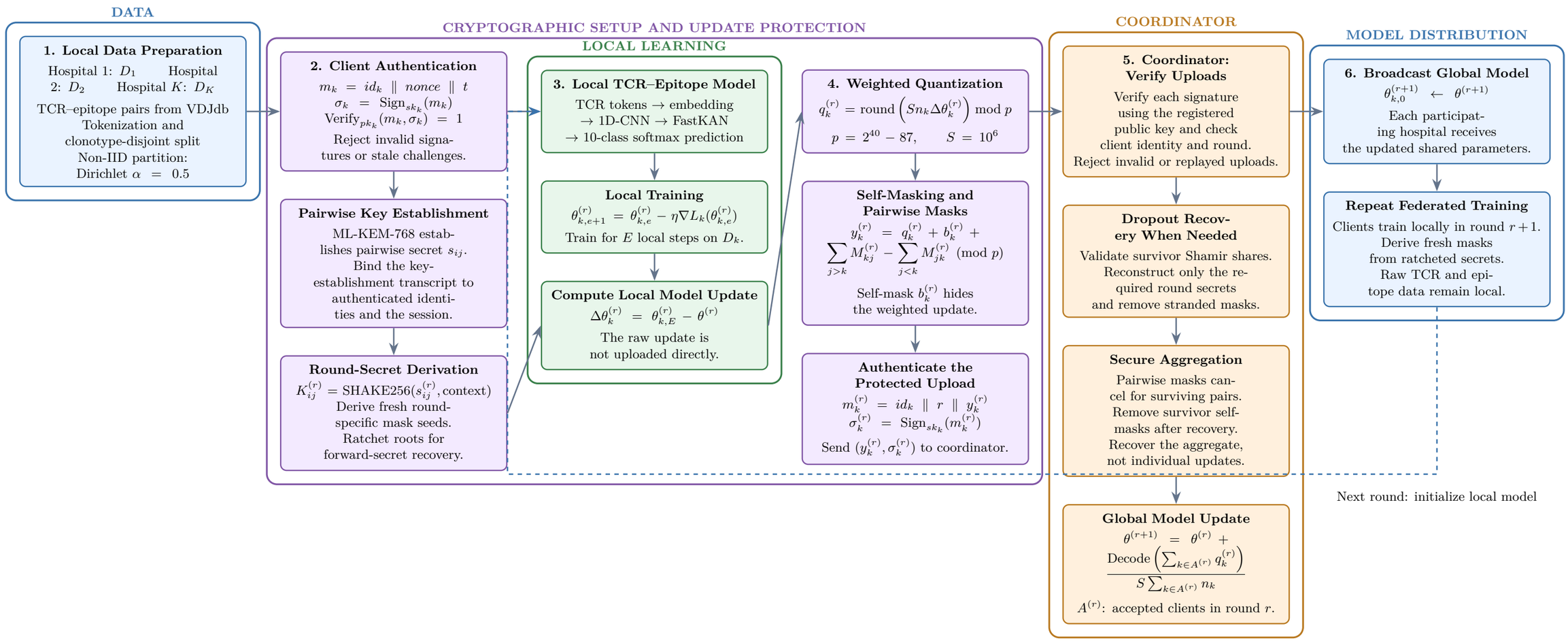
RodentNet: Privacy-Preserving Federated Learning for Collaborative Immunology

Architecture, Cryptographic Protection, and Federated Optimization

Raw TCR–epitope records remain local; authenticated masked model updates support collaborative learning.



Equation Legend

Federated objective

$$F(\Theta) = \sum_{k=1}^{K} \alpha_k F_k(\Theta), \qquad \alpha_k = \frac{n_k}{\sum_j n_j}.$$

Institutions are weighted by their number of training examples.

Pairwise mask generation

$$M_{ij}^{(r)} = \mathrm{SHAKE256}(\mathrm{context}, i, j, r, s_{ij}^{(r)}) \bmod p.$$

The same pairwise mask is added at one client and subtracted at the other.

Threshold dropout recovery

$$f(z) = s + \sum_{\ell=1}^{t-1} a_\ell z^\ell \pmod q, \qquad share_i = (i, f(i)).$$

At least t valid shares reconstruct the secret using Lagrange interpolation.

Security scope: ML-DSA-65 provides digital signatures; ML-KEM-768 establishes shared secrets. Secure aggregation hides individual updates under the stated protocol assumptions. It does not, by itself, prevent data poisoning or leakage from the final aggregate.